

Mid-West University
Graduate School of Engineering



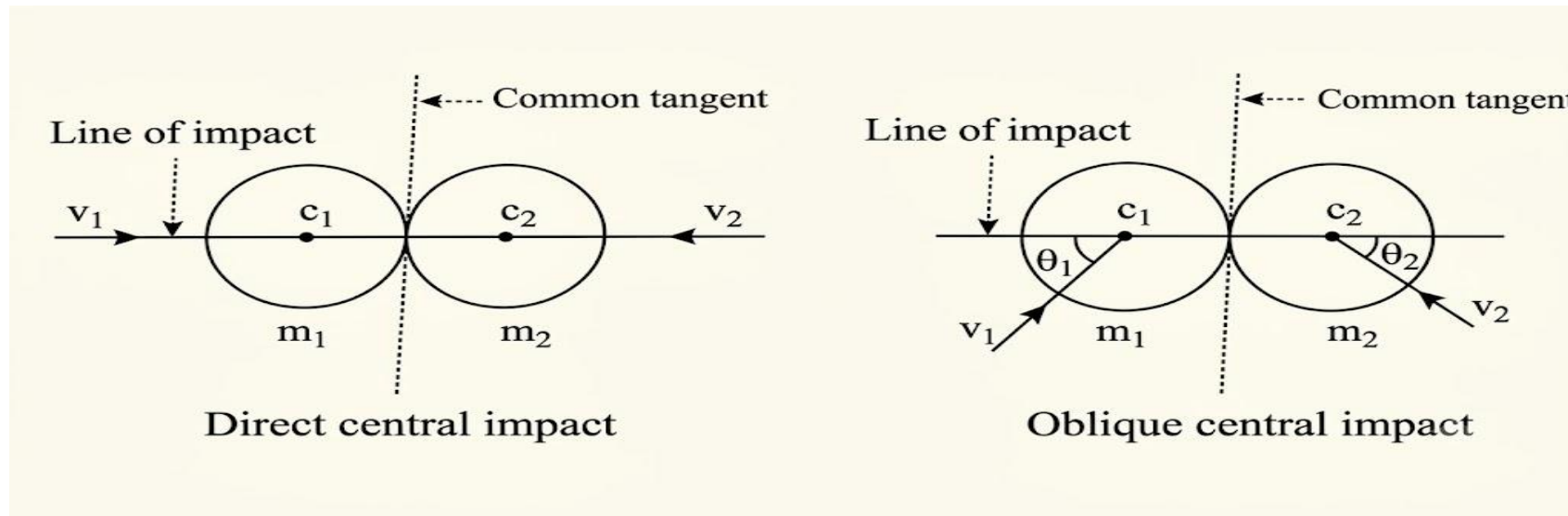
Chapter :Impact

Lecture Slides

By:
Amit Paudel

Impact

A collision between two bodies which occurs in a very short interval of time and during which the two bodies exert relatively large forces on each other is called an **impact**. The common normal to the surface in contact during the impact is called **Line of Impact**. If the mass centres of the two colliding bodies are located on this line of impact, the impact is called **Central Impact** otherwise **Eccentric Impact**. If the velocities of the two particles are directed along the line of impact, the impact is said to be **Direct Impact**. If either or both particles move along a line other than the line of impact, it is said to be **Oblique Impact**.



Direct Central Impact

- Consider two particles A and B of mass m_A and m_B , which are moving in the same straight line with known velocities $v_A > v_B$; particle A will eventually strike particle B.
- Under the impact the particles will deform and at the end of period of deformation, they will have common (same) velocity u .
- A period of restitution (return back to original position) will take place depending upon the material involved, two particles either will have regained their original shape or will stay permanently deformed.

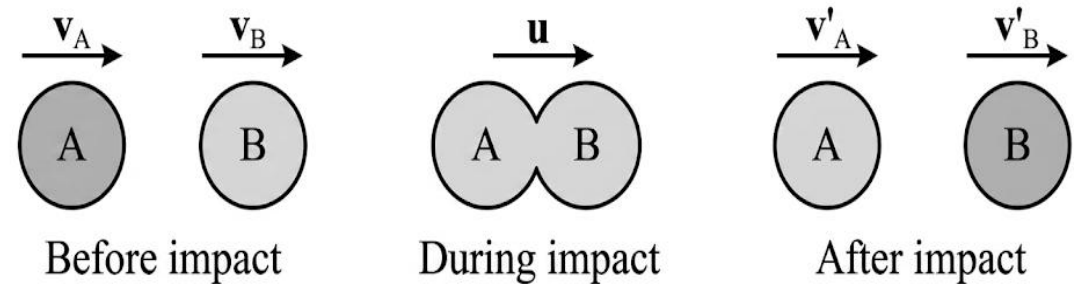
Considering first two particles as a single system, we note that there is no impulsive external force, so

$$m_A \bar{v}_A + m_B \bar{v}_B = m_A \bar{v}'_A + m_B \bar{v}'_B$$

since all velocities are directed along x-axis

$$m_A v_A + m_B v_B = m_A v'_A + m_B v'_B$$

($\rightarrow$ +ve direction) and ($\leftarrow$ -ve direction)



Using the definition of coefficient of restitution,

$$e = \frac{\text{Relative velocity after impact}}{\text{Relative velocity before impact}}$$

$$= \frac{v'_B - v'_A}{v_A - v_B}$$

$$\therefore = \frac{v'_B - v'_A}{v_A - v_B}$$

$$v'_B - v'_A = e (v_A - v_B)$$

Cases:

(i) Perfectly plastic impact ($e = 0$)

$$e = \frac{v'_B - v'_A}{v_A - v_B}$$

$$0 = v'_B - v'_A$$

$\therefore v'_A = v'_B = v_c$ is common velocity.

$$\text{Loss of K.E} = \left(\frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 \right) - \frac{1}{2} (m_A + m_B) \cdot v_c^2$$

Momentum equation;

$$m_A v_A + m_B v_B = (m_A + m_B) \cdot v_c$$

(ii) Perfectly elastic impact ($e = 1$)

$$e = \frac{v'_B - v'_A}{v_A - v_B}$$

$$v_A - v_B = v'_B - v'_A \quad (\text{for, } e = 1)$$

$$v_A + v'_A = v_B + v'_B \dots\dots\dots(i)$$

Prove that kinetic energy is conserved in perfectly elastic collision:

Since no external forces are involved , from linear momentum equation;

$$\mathbf{m_A v_A + m_B v_B = m_A v'_A + m_B v'_B}$$

$$\mathbf{m_A(v_A - v'_A) = m_B(v'_B - v_B)(ii)}$$

multiply eq. (I) by eq. (II)

$$\mathbf{m_A(v_A - v'_A)(v_A + v'_A) = m_B(v'_B - v_B)(v_B + v'_B)}$$

$$\mathbf{m_A(v_A^2 - v'^2_A) = m_B(v'^2_B - v^2_B)}$$

$$\frac{1}{2} m_A v_A^2 - \frac{1}{2} m_A v_A'^2 = \frac{1}{2} m_B v_B'^2 - \frac{1}{2} m_B v_B^2$$

$$\frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 = \frac{1}{2} m_A v_A'^2 + \frac{1}{2} m_B v_B'^2$$

Initial KE = Final KE

So, kinetic Energy of particle is conserved during elastic impact.

If $0 < e < 1$: loss of KE = (KE)_{before impact} - (KE)_{after impact}

$$= \left[\frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 \right] - \left[\frac{1}{2} m_A v_A'^2 + \frac{1}{2} m_B v_B'^2 \right]$$

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Chapter7: Mechanical Vibration

Lecture Slides

By:
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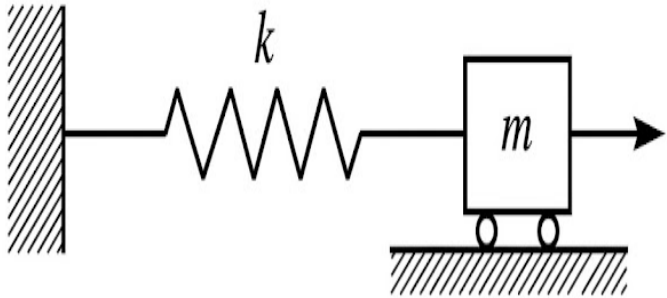
Mechanical Vibration

7.1 Introduction

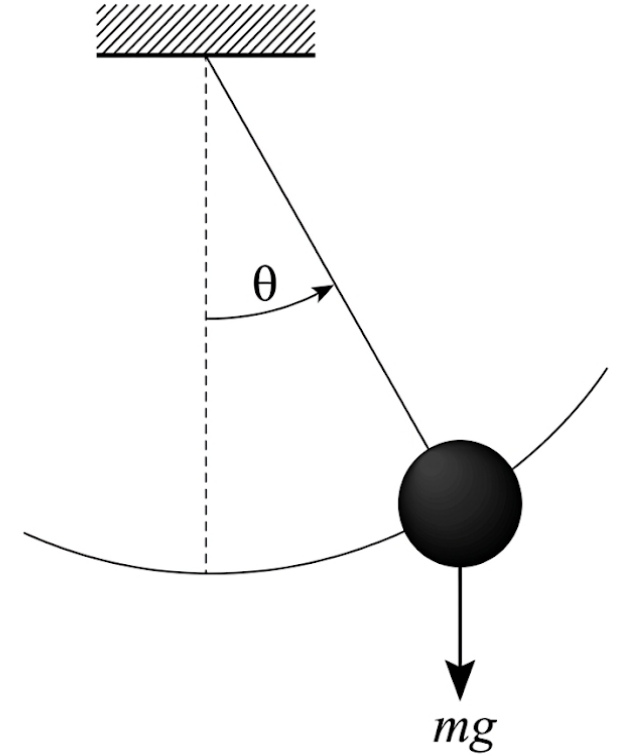
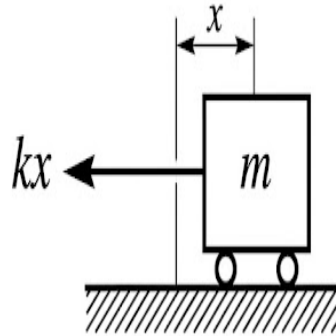
- A mechanical vibration is the motion of a particle or a body which oscillates about a position of equilibrium.
- The mechanical vibration generally results when a system is displaced from a position of stable equilibrium.
- The system tends to return to this position under the action of restoring forces.
- But the system generally reaches its original position with a certain acquired velocity which carries it beyond that position.
- Since the process can be repeated indefinitely, the system keeps moving back and forth across its position of equilibrium.

Mechanical Vibration

A: Spring-Mass System in Initial Position



B: Free Body Diagram - Displaced State



Introduction

Vibration can be defined as a to-and-fro motion of a body or a system about its equilibrium position.

Vibration is sometimes also called an oscillation.

Any body or system having inertia and elasticity is capable of vibration.

Vibration is initiated when a body having inertia is displaced from its equilibrium position due to some external disturbances.

A restoring or conservative force is developed due to elasticity of the system pulls the system back towards its equilibrium and to-and-fro motion continues if there is not any dissipative force.

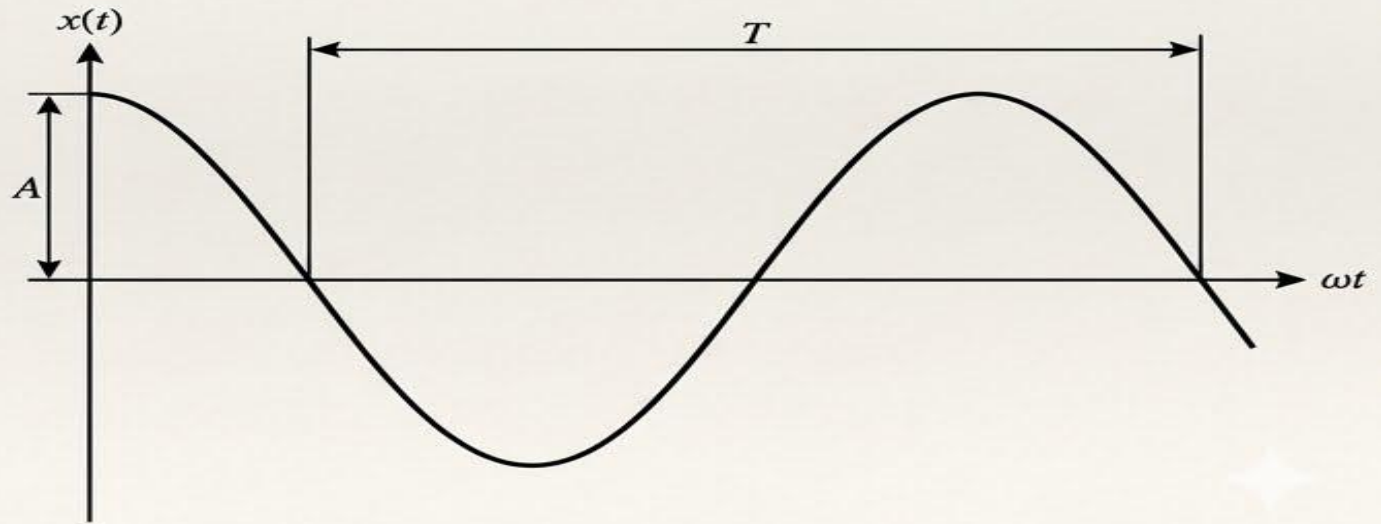
- The time interval required for a system to complete a full cycle of motion is called period of vibration(T). unit is **second**
- The number of cycle per unit time defines the frequency (f) = $1/T$. unit is **hertz**. ($1/s$).
- The maximum displacement of the system from its position of equilibrium is called **amplitude** of the vibration.
- When the motion is maintained by the restoring force only, the vibration is said to be **free vibration**.
- When the periodic force is applied, the resulting motion is **forced vibration**.

Simple Harmonic Motion

The simplest form of oscillatory or vibratory motion is periodic motion. Any motion that repeats itself after a constant time interval is called periodic motion. A most common example of periodic motion is simple harmonic motion.

The instantaneous position of a particle undergoing simple harmonic motion can be represented by circular functions, specifically sine or cosine.

$$x(t) = A \cos(\omega t + \phi)$$



Since the circular function repeats itself in 2π radians and take a time interval of T ($\omega T = 2\pi$), then

$$T = \frac{2\pi}{\omega}$$

where T is called time period of the cycle measured in s and ω is the circular frequency of the oscillation measured in rad/s .

Similarly, frequency of oscillation f which is the number of cycles measured in Hz can be expressed as

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

Types of Vibrations:

Free Vibration : Vibration only due to restoring force like elastic force.

Forced Vibration : Vibration due to external periodic force.

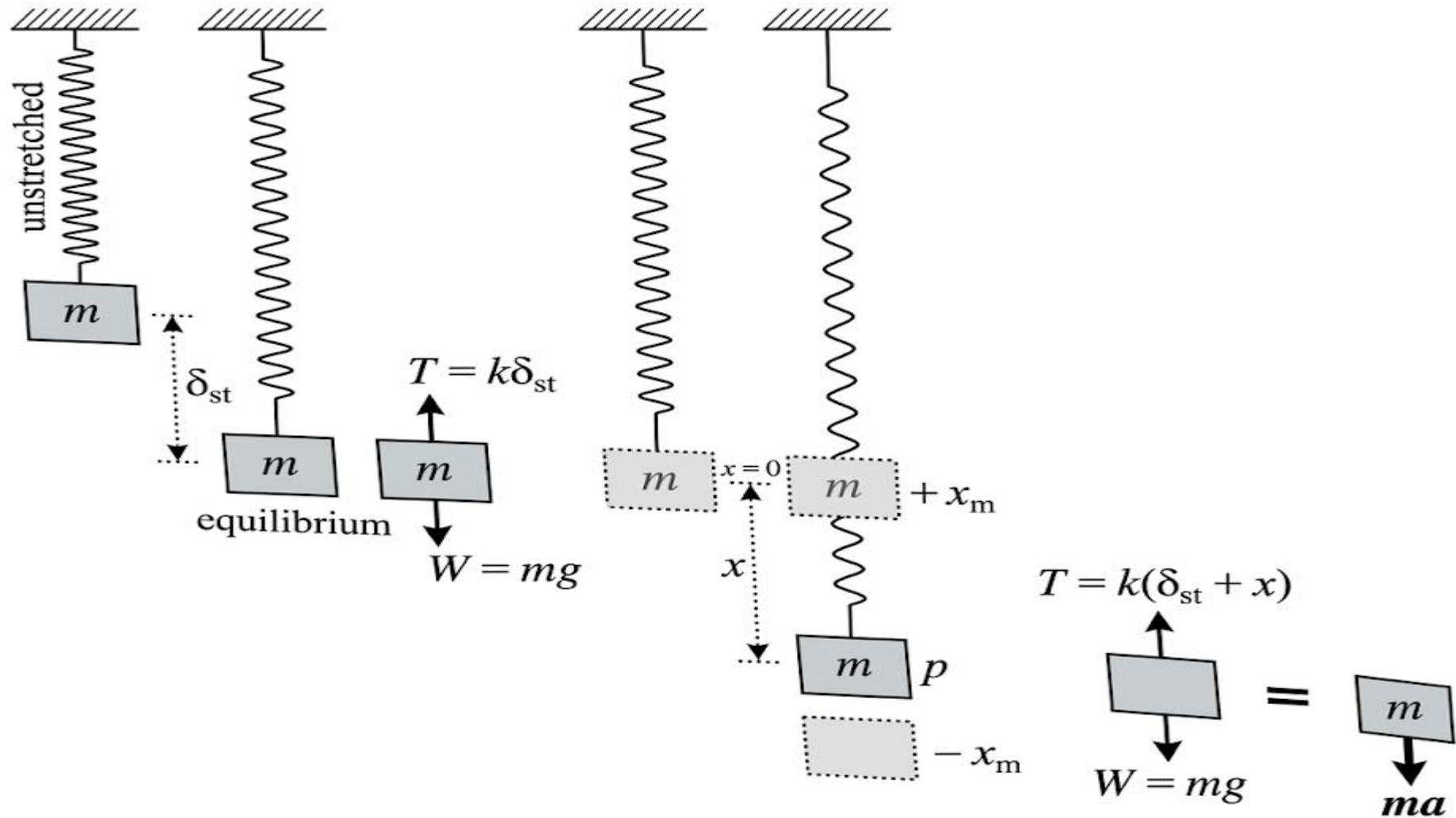
Un-damped Vibration : Effect of friction may be neglected

Damped Vibration : Effect of friction is considered ; the vibration slowly decreases and comes to halt

Key Applications in Civil Engineering

- Earthquake and Wind Resistance:** Engineers analyze natural frequencies to design structures that withstand seismic activity and high winds without collapsing.
- Vibration Control Devices:** Systems like tuned mass dampers are installed in tall skyscrapers and long-span bridges to absorb kinetic energy and reduce swaying.
- Structural Health Monitoring (SHM):** Sensors measure ambient vibrations in real time to detect hidden cracks, structural fatigue, or loss of rigidity in bridges and dams.
- Non-Destructive Testing (NDT):** Controlled vibrations help evaluate the integrity of concrete piles, foundations, and soil layers without damaging the structure.
- Construction Equipment:** Heavy machinery like vibratory rollers use mechanical oscillation to compact soil, asphalt, and concrete effectively on site.

Un-damped free vibration of particles



Suppose now that the particle is displaced through a distance of x m from its equilibrium position and released with no initial velocity. Now the particle will move back and forth through its equilibrium position. To analyze the vibration, let us consider the particle in a position P at some arbitrary time t . Denoting the displacement by ' x '. We note that forces acting on the particle are weight ' w ' and spring force T .

If F be the magnitude of resultant forces;

$$F = W - k(\delta_{st} + x)$$

$$= W - k\delta_{st} - kx$$

$$F = -kx$$

Substituting the fundamental equation $F = ma$.

$$ma = -kx$$

$$m\ddot{x} + kx = 0.$$

$$\frac{md^2x}{dt^2} + kx = 0$$

or, $\frac{d^2x}{dt^2} + \frac{k}{m} \cdot x = 0$

$$\frac{d^2x}{dt^2} + \omega_n^2 x = 0$$

Where ω_n is the natural circular frequency;

solving this differential equation;

$$x = x_m \sin(\omega_n t + \phi)$$

Where ϕ is phase angle.

$$\therefore x = x_m \sin(\omega_n t + \phi)$$

$$v = \dot{x} = x_m \omega_n \cos(\omega_n t + \phi)$$

$$a = \ddot{x} = -x_m \omega_n^2 \sin(\omega_n t + \phi).$$

$$\text{Natural frequency (f)} = \frac{\omega_n}{2\pi} \text{ (Hz)}$$

$$\text{Period (T)} = \frac{2\pi}{\omega_n} = \frac{1}{f} \text{ (sec)}$$

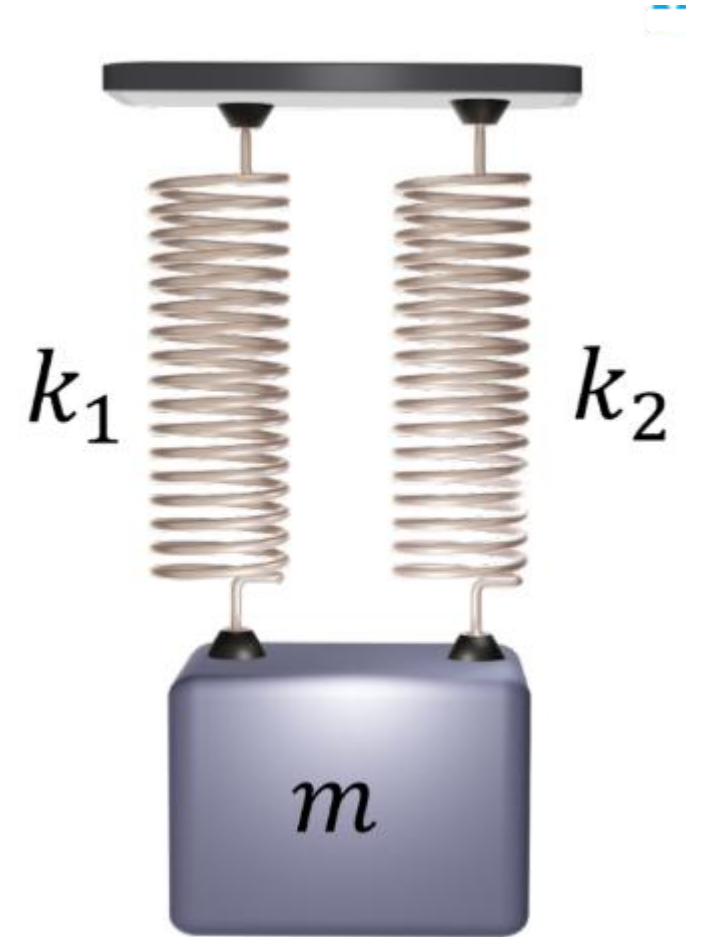
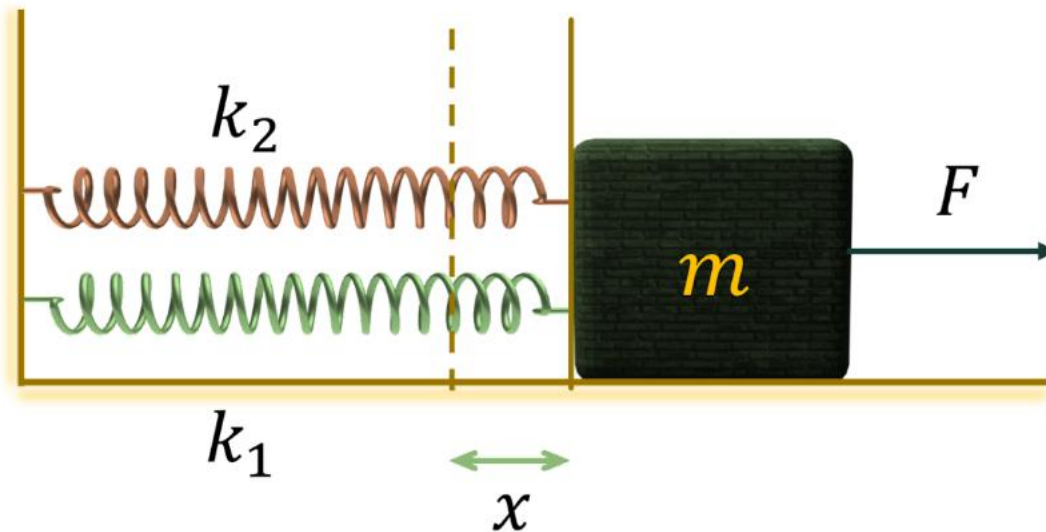
From above equation, we note that

$$v_{\max} = x_m \omega_n$$

$$a_{\max} = x_m \omega_n^2$$

Parallel combination

The parallel combinations of springs are shown in the figure. If the parallel combination of two massless springs is stretched up to a distance by some external force, then the extension of both the springs will be the same.



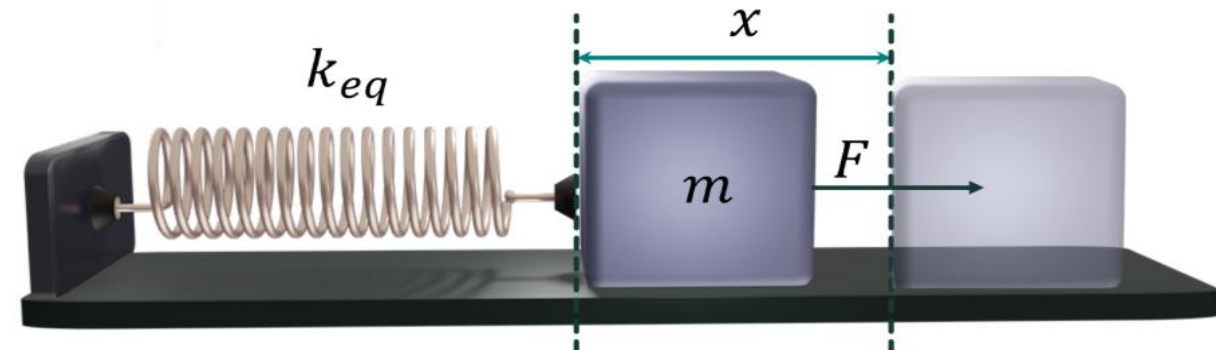
Suppose a force F is applied to the block then it will distribute on both the springs. let the individual springs of spring constants k_1 and k_2 feel the force F_1 and F_2 , respectively.

Then,

$$F = F_1 + F_2$$

Where $F_1 = k_1 x$ and $F_2 = k_2 x$

Hence, $F = k_1 x + k_2 x \dots\dots (i)$



Now, suppose that the whole system is replaced by a single spring of spring constant k_{eq} and the same amount of extension occurs due to the same force F as shown in the figure.

Then

$$F = k_{eq} x$$

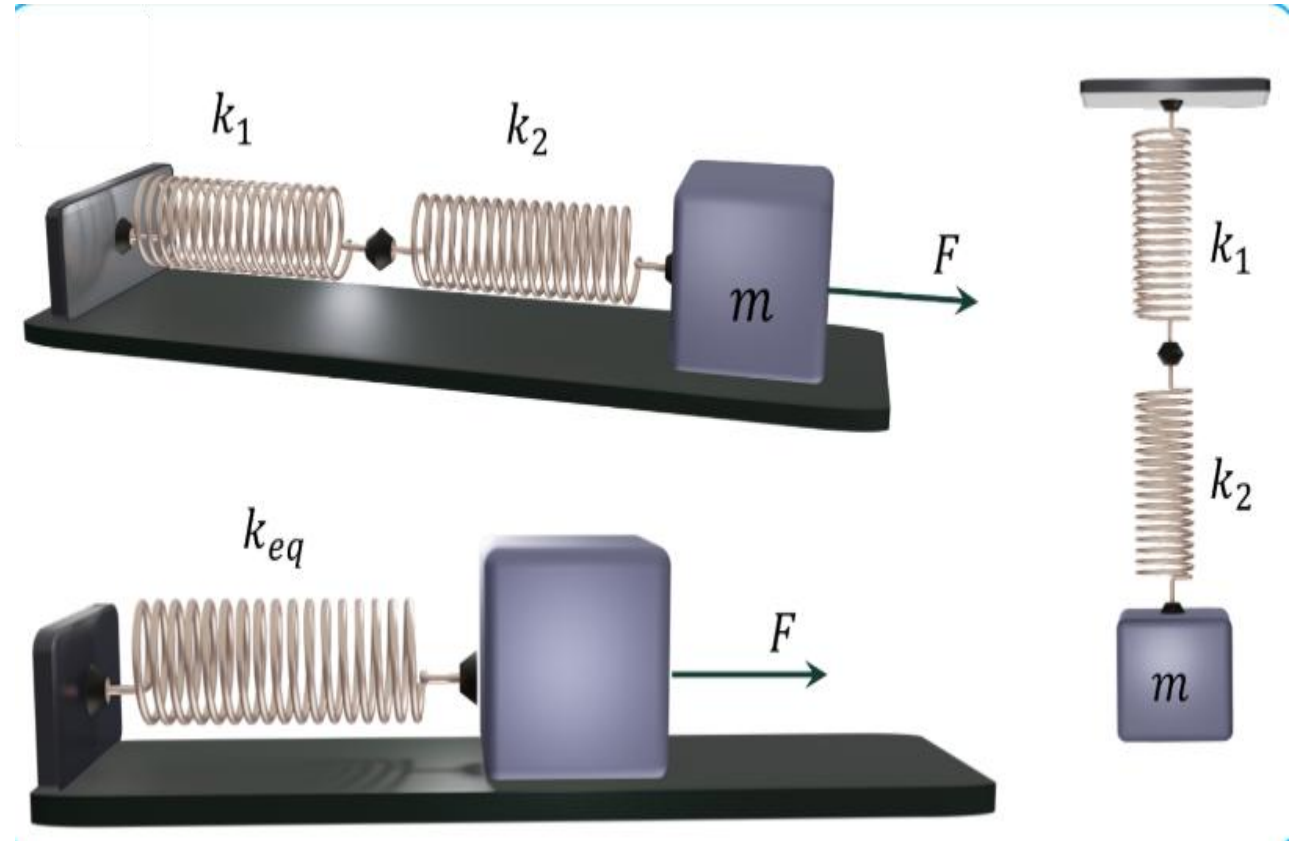
From eq(i)

$$k_{eq} x = k_1 x + k_2 x$$

$$k_{eq} = k_1 + k_2$$

Series combination

- The series combination of springs is shown in figures below.
- In a series combination of springs, the extension in the springs is different for the force of the same magnitude.
- The extension in the springs depends on the stiffness.



For the same force F , the extensions will be x_1 and x_2 having spring constants k_1 and k_2 .

Then

$$F = k_1 x_1 \text{ and } F = k_2 x_2 \dots\dots\dots (i)$$

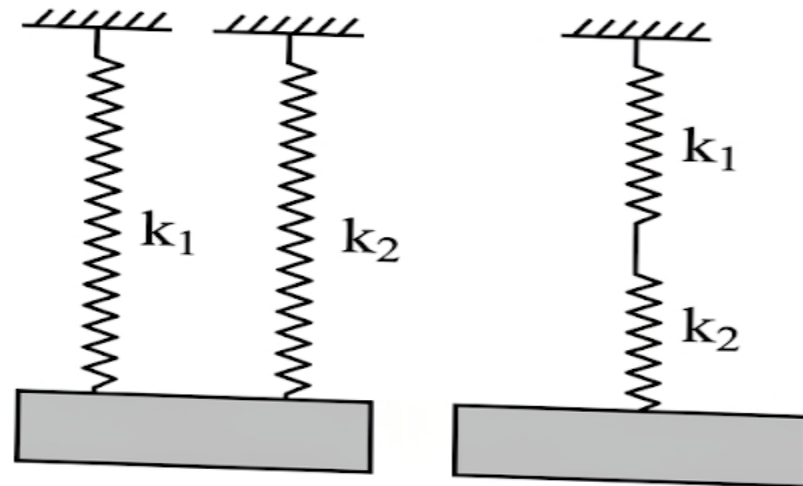
Suppose that the whole system is replaced by a single spring of spring constant k_{eq} and the net extension x occurs as shown in the figure. Then As total extension x will be equal to the sum of extension of individual spring i.e.

$$x = x_1 + x_2$$

Substitute the values of x, x_1 and x_2 from equation (i) and (ii), we get

$$\begin{aligned} \frac{F}{k_{eq}} &= \frac{F}{k_1} + \frac{F}{k_2} \\ \frac{1}{k_{eq}} &= \frac{1}{k_1} + \frac{1}{k_2} \end{aligned}$$

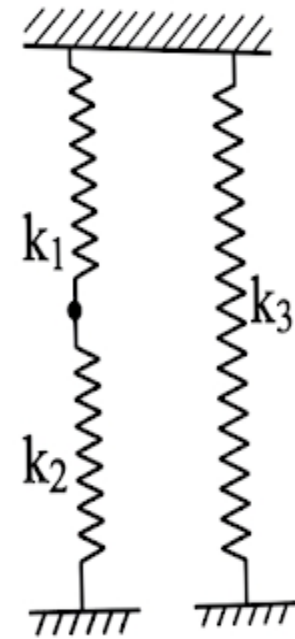
1. A 50 kg block moves as shown in figure. The block is pulled 50mm down from its equilibrium position and released for each spring arrangement, determine
- (a) The period of vibration.
 - (b) The maximum velocity of the block.
 - (c) The maximum acceleration of the block.



3. A 13.6kg block is supported by the spring arrangement as shown. If the block is moved from its equilibrium position 44mm vertically downward and released, determine:

Take $K_1 = 3.5\text{kN/m}$, $K_2 = 2.1\text{kN/m}$ and $K_3 = 2.8\text{kN/m}$

- (a) The period and frequency.
- (b) The maximum velocity & maximum acceleration.



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Chapter5: Kinematics of Rigid Body

Lecture Slides

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Kinematics of Rigid Bodies

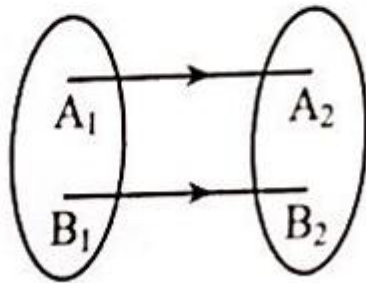
In this chapter, we will investigate the relations existing between the time the positions, the velocities and the accelerations of the various forming a rigid body.

Various types of rigid-body motion:

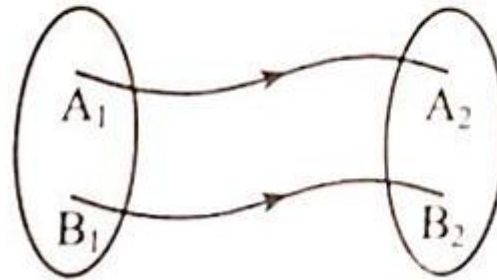
Various types of rigid body motion can be conveniently grouped as follows:

(1) Translation:

A motion is said to be translation if the body keeps same orientation during the motion. All the particles forming the body moves along parallel paths and if these path are straight line, the motion is said to be rectilinear translation and if these path are curved lines, the motion is a curvilinear translation.



Rectilinear translation

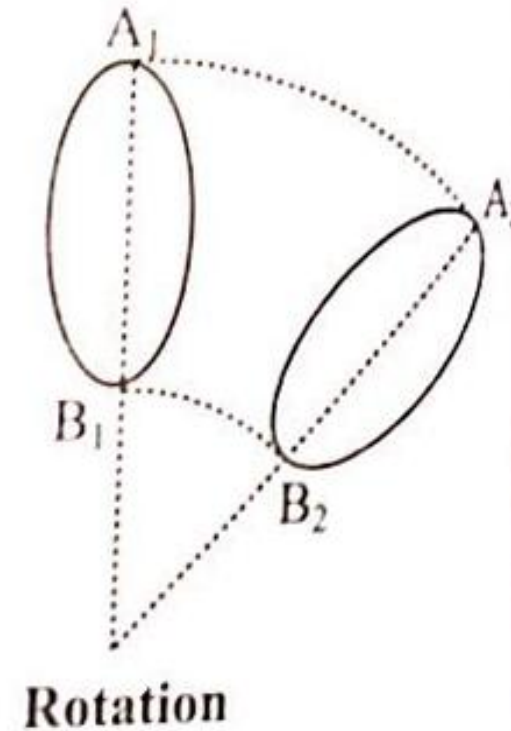


Curvilinear translation

(2) Rotation about fixed Axis:

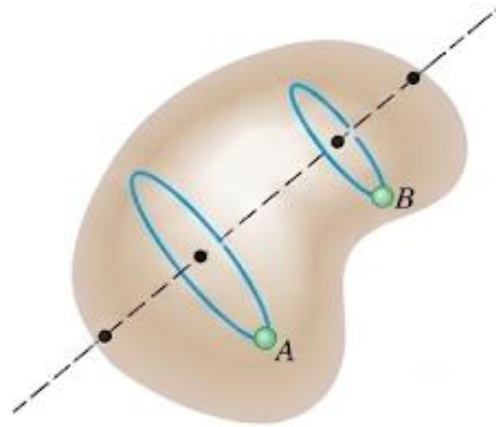
In this motion, the particles forming a rigid body moves in a parallel plane along circle centred on the same fixed axis i.e. particle moves along a concentric circle.

Rotation should not be confused with certain types of curvilinear translation.

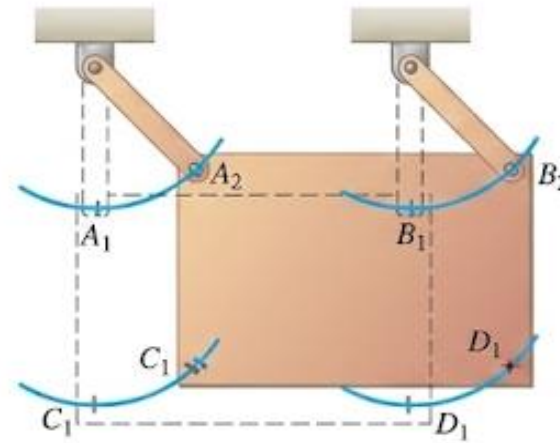


Rotation

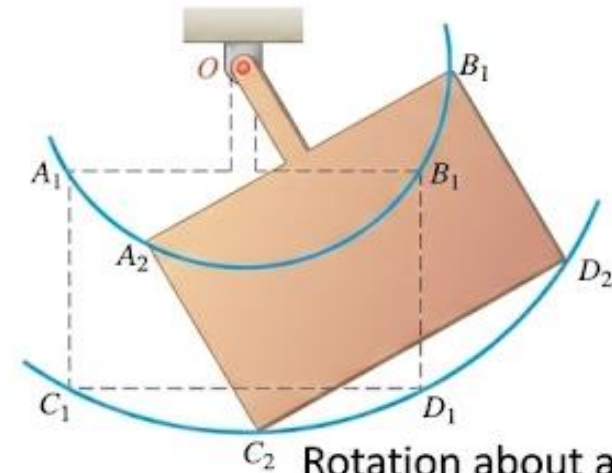
- A motion is said to be in rotation when the particles of the representative slab move along concentric circles.
- In curvilinear translation, particles move along parallel circles. In rotation, particles move along concentric circles.



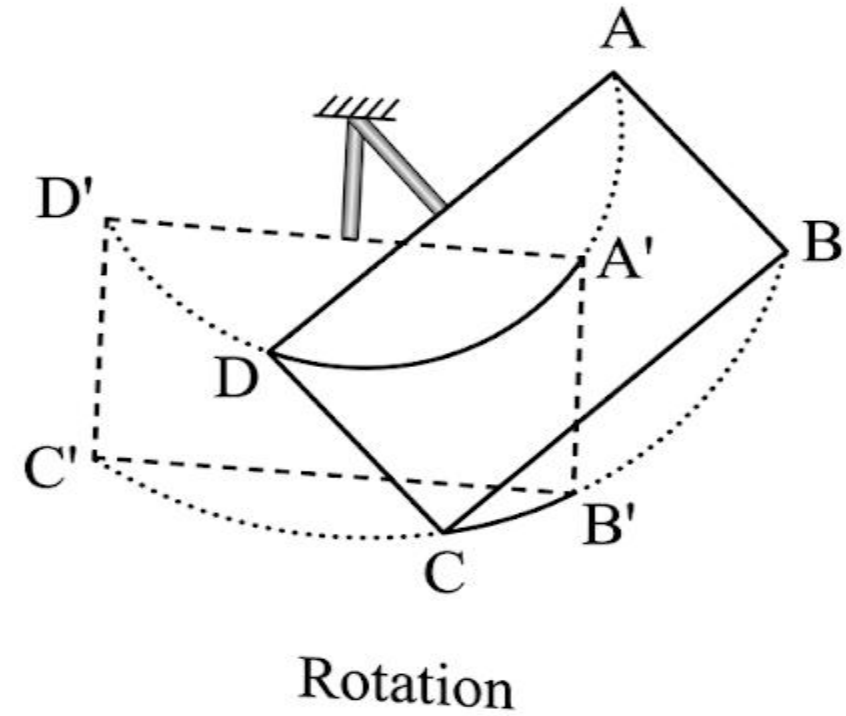
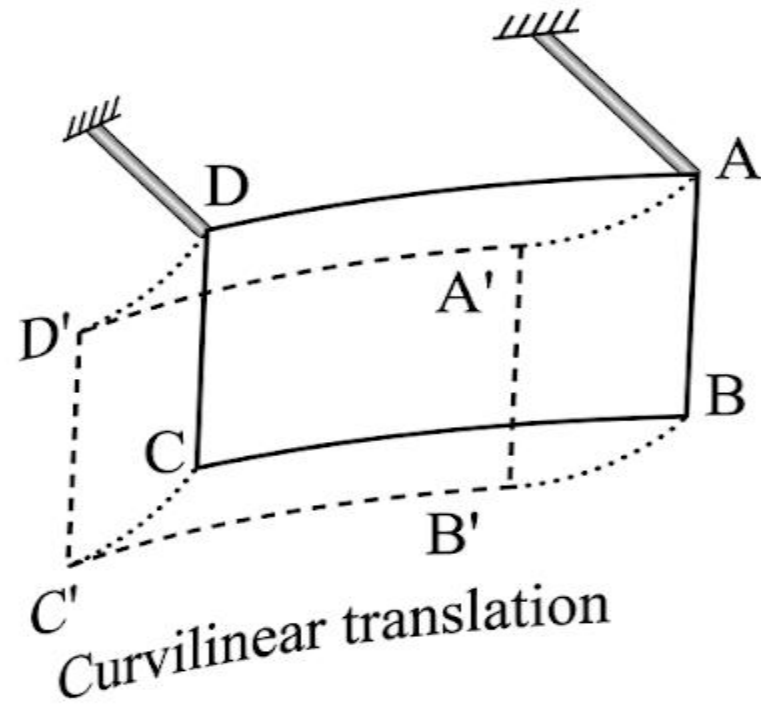
Rotation about an axis



(a) Curvilinear translation



(b) Rotation about a point



Note: In curvilinear translation particle moves along parallel circle while in rotation the particle moves along concentric circle.

Review of Rotary Motion

$$1 \text{ revolution} = 2\pi^c = 360^\circ$$

$$\text{Angular distance } (d\theta) = \frac{l}{r}$$

$$\text{Angular velocity } (\omega) = \frac{d\theta}{dt.}$$

$$\begin{aligned} \text{Angular acceleration } (\alpha) &= \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} \\ &= \frac{d\omega}{d\theta / \omega} \end{aligned}$$



Unit of $d\theta$ is rad.

Unit of ω is rad/s.

Unit of α is rad/s².

$$1 \text{ rpm} = 1 \times \frac{2\pi}{60}$$

$$N \text{ rpm} = \frac{2\pi}{60} \times N \cdot \text{rpm}$$

For uniform rotation;

$$\omega = \text{constant}$$

$$\square \quad \alpha = 0.$$

For uniform accelerated motion; $\alpha = \text{constant}$.

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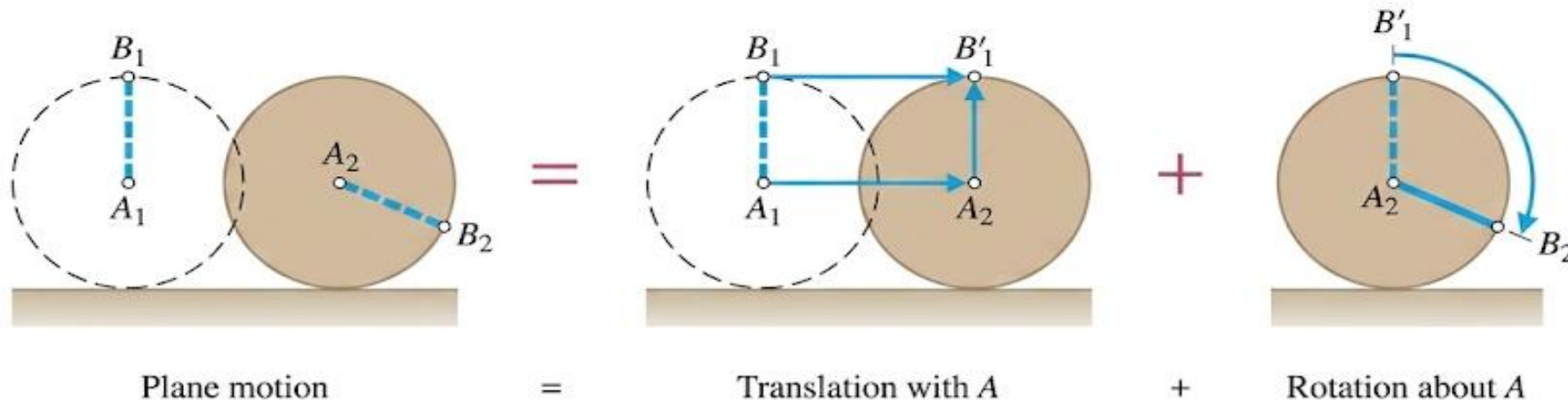
Chapter5: Kinetics of Rigid Body

Lecture Slides

By:
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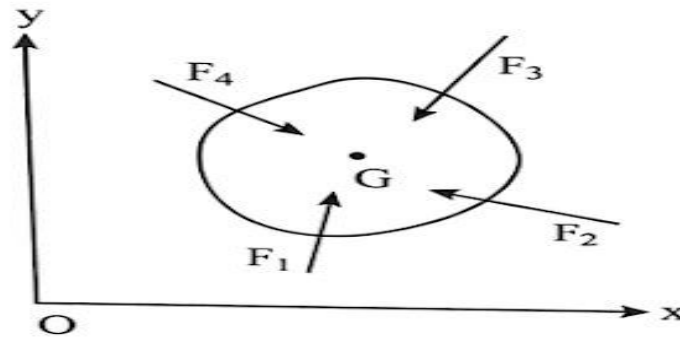
General Plane Motion

- General plane motion is a plane motion which is neither a translation nor a rotation. However, a general plane motion can always be considered as the sum of a translation and a rotation.
- Illustrations:



D'ALEMBERT'S PRINCIPLE

Consider a rigid slab of mass m moving under the action of several external forces $\vec{F}_1, \vec{F}_2, \vec{F}_3, \dots$ contained in the plane of the slab as shown in figure.

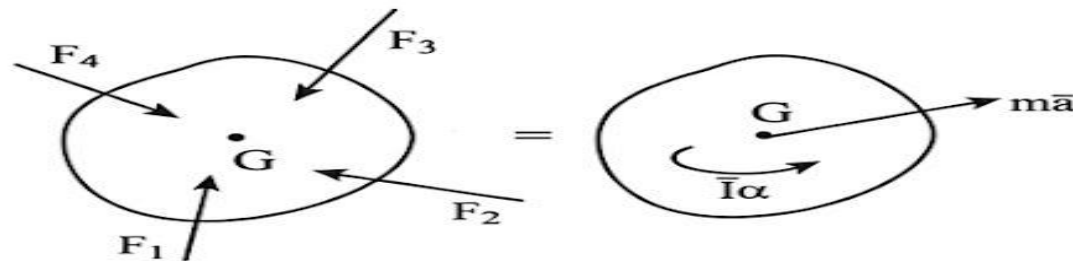


$$\Sigma F_x = m\bar{a}_x$$

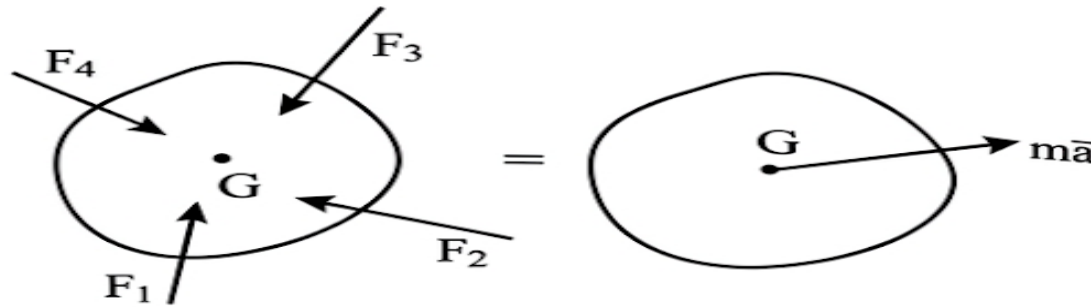
$$\Sigma F_y = m\bar{a}_y$$

$$\Sigma M_G = I\bar{\alpha}$$

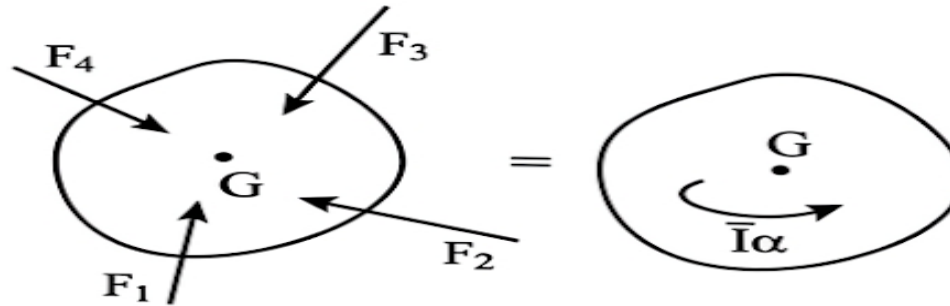
“The external forces acting on a rigid body are equivalent to the effective force of the various particles forming the body.” This is called **D'ALEMBERT'S PRINCIPLE**.



- (a) **Translation:** In the case of translation the angular acceleration of the body is $\alpha = 0$, & effective forces reduces to $m\bar{a}$ attached at G.
 $(\Sigma F)_{\text{eff}} = m\bar{a}$



- (b) **Centroidal rotation:** When the body is in centroidal rotation, $\bar{a}$ reduces to zero, the effective forces of the body reduces to the couple $\bar{I}\alpha$.
 $(\Sigma F)_{\text{eff}} = \bar{I}\alpha$.



Principle of work and Energy for a rigid body:

In order to apply the principle of work and energy to the analysis of the motion of rigid body, it is assumed that the rigid body is made of a large number 'n' of the particles of mass Δm_i .

$$T_1 + U_{1 \rightarrow 2} = T_2$$

T_1 & T_2 initial and final values of total kinetic energy of particles forming the rigid body.

$U_{1 \rightarrow 2}$ work of all forces acting on various particles of body.

$$T = \frac{1}{2} \sum_{i=1}^n \Delta m_i v_i^2$$

Principle of work and Energy for a rigid body:

a. Work of forces acting on a rigid body.

The work of a force F during a displacement of its point of application from A_1 to A_2 is

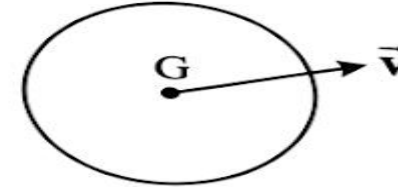
$$U_{1 \rightarrow 2} = \int_{A_1}^{A_2} \vec{F} \cdot d\vec{r}$$

$$U_{1 \rightarrow 2} = \int_{S_1}^{S_2} (F \cos \alpha \cdot ds)$$

b. Kinetic energy of a rigid body.

(i) Kinetic energy in translation:

Consider a rigid body in translation since at a given instant, all particles of the body have the same velocity as the mass centre i.e $\mathbf{v} = \bar{\mathbf{v}}$.



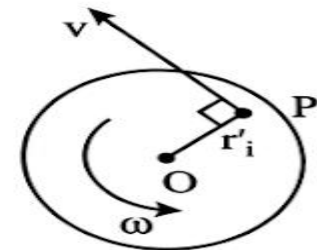
$$KE = T = \sum_{i=1}^n \frac{1}{2} (\Delta m_i) \bar{v}^2$$

$$T = \frac{1}{2} m \bar{v}^2$$

(b) Kinetic energy in rotation:

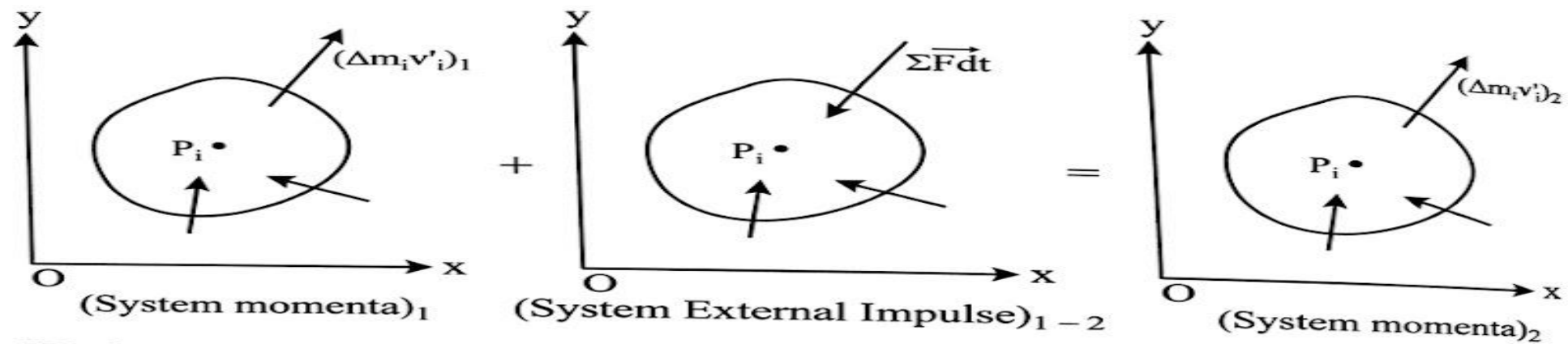
Consider a rigid body rotating about a fixed axis.

$$\begin{aligned} KE = T &= \sum \frac{1}{2} (\Delta m) v^2 \\ &= \sum \frac{1}{2} \Delta m. (\omega r)^2 \\ &= \sum \frac{1}{2} \Delta m r^2. \omega^2 \\ &= \frac{1}{2} \omega^2 \sum r^2 \Delta m. \\ &= \frac{1}{2} I_o \omega^2 (I = m r^2). \end{aligned}$$



Principle of Impulse and Momentum for the plane motion of a rigid body.

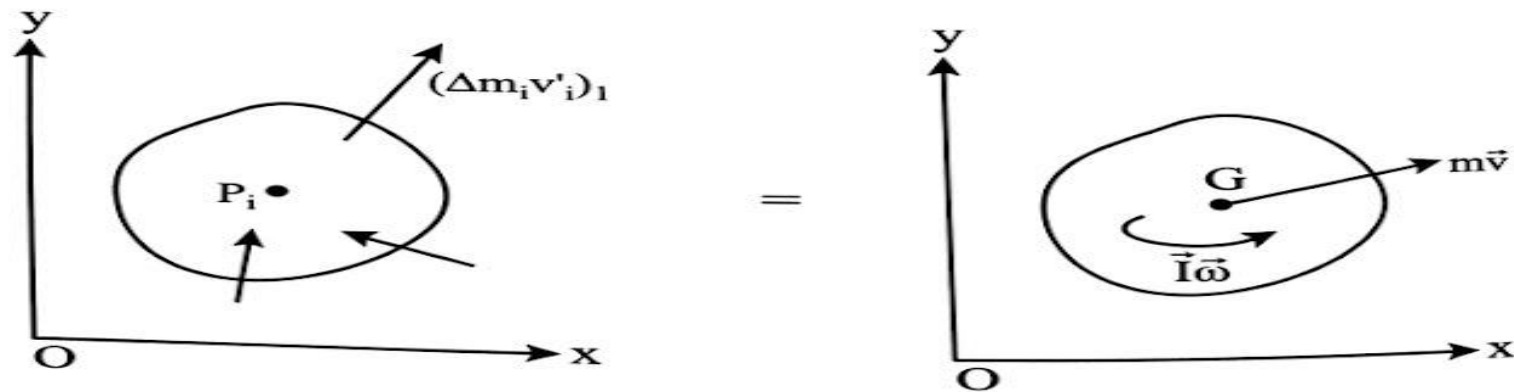
Consider a rigid body made of a large number of particles P_i .



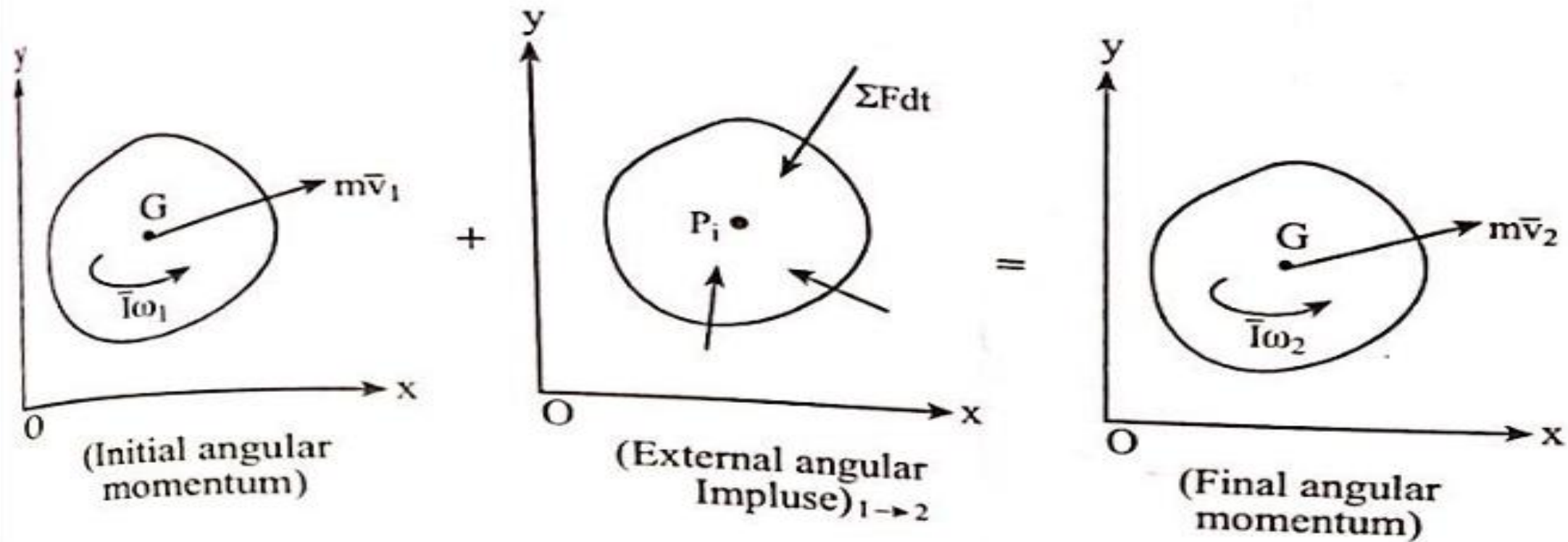
We know;

$$\vec{L} = \sum_{i=1}^n \vec{v}_i \Delta m_i$$

$$\vec{H}_G = \sum_{i=1}^n \vec{r}'_i \times \vec{v}_i \Delta m_i$$

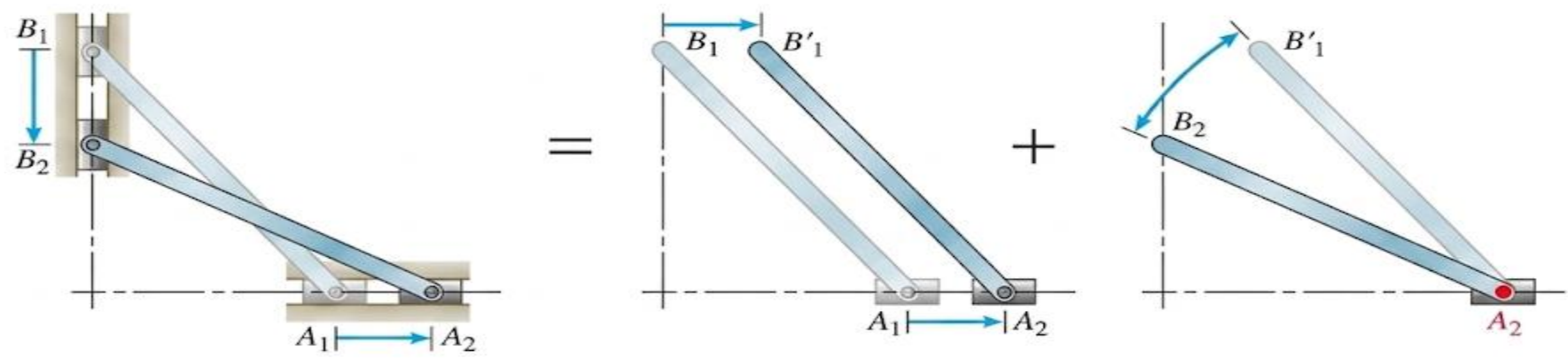


So, in case of plane motion,



Conservation of a angular momentum.

$$(H_o)_1 = (H_o)_2.$$



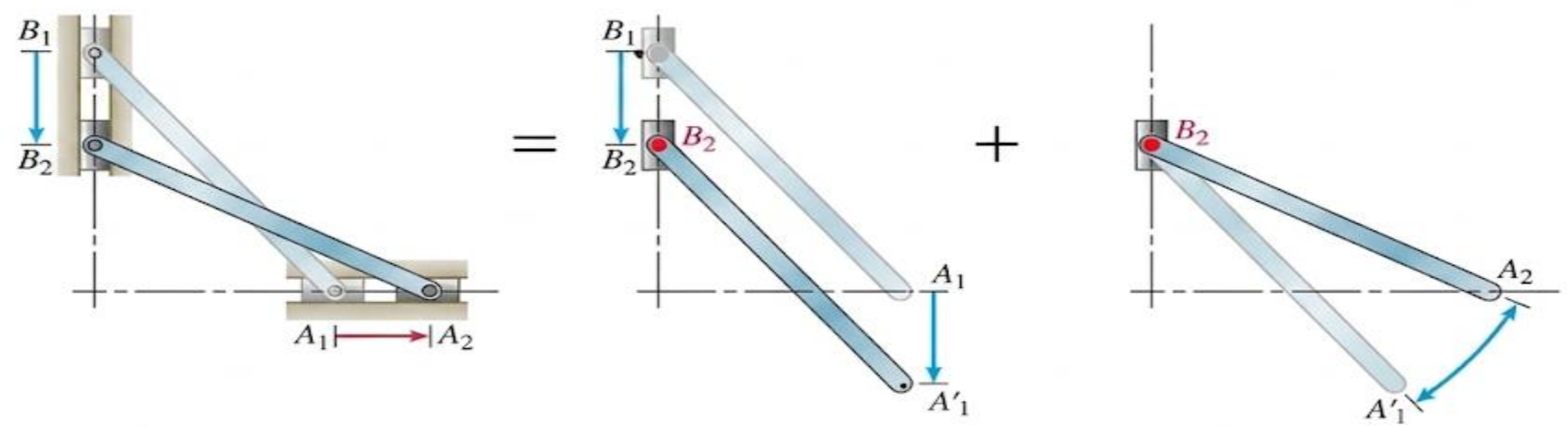
Plane motion

=

Translation with A

+

Rotation about A



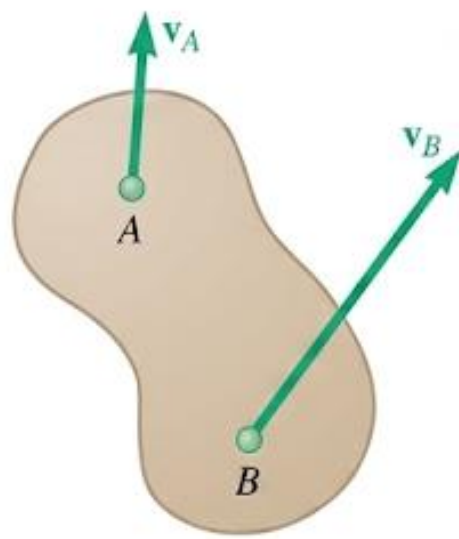
Plane motion

=

Translation with B

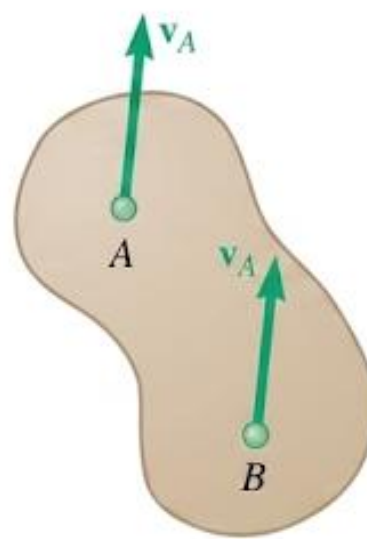
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Rotation about B



Plane motion

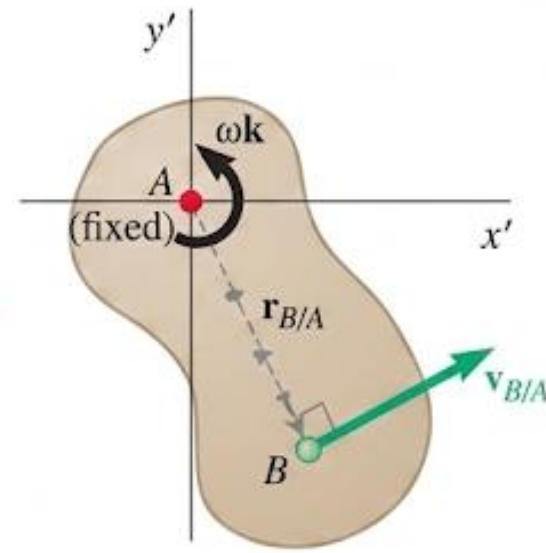
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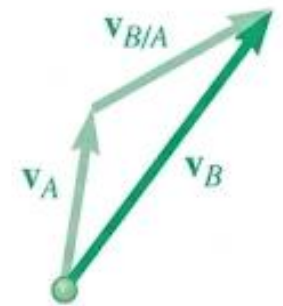
Translation with A

+

$$\mathbf{v}_{B/A} = \boldsymbol{\omega} \mathbf{k} \times \mathbf{r}_{B/A} \quad \mathbf{v}_{B/A} = r\boldsymbol{\omega}$$



Rotation about A



$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$

Absolute and Relative Velocity in Plane Motion

- We saw in the preceding section that any plane motion of a slab can be replaced by a translation defined by the motion of an arbitrary reference point A and a simultaneous rotation about A . The absolute velocity $\mathbf{v_B}$ of a particle B of the slab is obtained from the relative velocity formula:

$$\mathbf{v_B} = \mathbf{v_A} + \mathbf{v_{B/A}}$$

Right-hand member represents a vector sum.

- The velocity $\mathbf{v_A}$ corresponds to the translation of the slab with A , while the relative velocity $\mathbf{v_{B/A}}$ is associated with the rotation of the slab about A and is measured with respect to axes centered at A and of fixed orientation.

Instantaneous Centre of Rotation

- Consider the general plane motion of a slab. We propose to show that “at any given instant the velocities of the various particles of the slab are the same as if the slab were rotating about a certain axis perpendicular to the plane of the slab”, called the *instantaneous axis of rotation*. This axis intersects the plane of the slab at a point C , called the *instantaneous center of rotation* of the slab.
- Plane motion of a slab can always be replaced by a translation defined by the motion of an arbitrary reference point A and by a rotation about A . As far as the velocities are concerned, the translation is characterized by the velocity $\mathbf{v}_A$ of the reference point A and the rotation is characterized by the angular velocity ω of the slab (which is independent of the choice of A). Thus, the velocity of point A , $\mathbf{v}_A$ and the angular velocity ω of the slab define completely the velocities of all the other particles of the slab.

